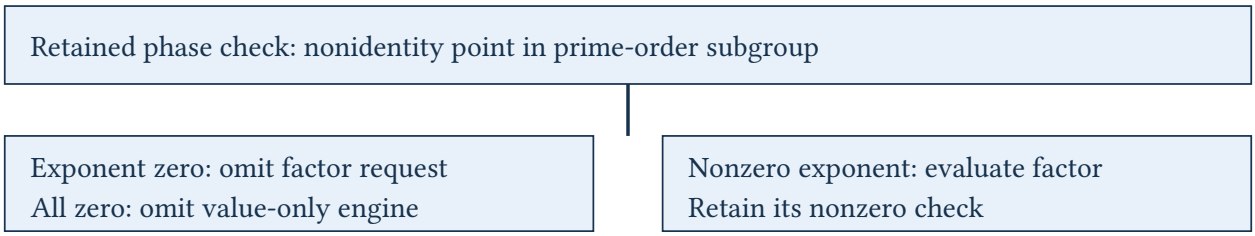
Omitting zero-exponent direct factors

2026-11-07 | PCC-20261107-7d2ab2 | Native development



Full-value sign validator, geometry and difference checks remain.

Domain argument. A nonidentity point of prime order r cannot be n-torsion for $0 < n < r$. Every selected division index lies in this interval; the normalized seed is nonzero. Hence a zero-exponent factor is nonzero and contributes one without evaluation. Primality remains a caller precondition. Phase and full-value checks are retained. [pcc:unverified]

Implementation. Skip zero exponents before division-value and seed-power requests. If every exponent is zero, return the unit pair without constructing a value-only division engine. Geometry lanes and coordinate-difference checks are unchanged.

Development evidence. Three selected native methods pass. Complete/split checks retain 499 subgroup rows, 100 splits, 1375 advances and 450 skips. Direct top-level factor requests are 1, 0 and 2 from 3, 4 and 4 available factors over F5/r7, F7/r13 and F509/r479. The all-zero case constructs no value-only engine. Claims are unsigned producer evidence; see receipt.json and test/limits/factors.log. [pcc:unverified]

Cost boundary. Phase computation still runs, and retained requests can recursively evaluate other indices. The paired full-value source remains mandatory. No scalar recovery, timing comparison or cloud experiment ran. The original cohort remains intact.

Next. Analyze retained seed dependence and projected geometry residuals, then measure complete work.

Full argument and exact fixtures: DERIVATION.md and inherited projected-transport report. Primary source: Sage division-polynomial documentation, https://doc.sagemath.org/html/en/reference/arithmetic_curves/sage/schemes/elliptic_curves/ell_generic.html. KB unavailable. No novelty-priority claim.